

📍 Connecticut, USA
☎ +1 (575) 339-8670
✉ tardivo.fabio@gmail.com
🌐 <https://fabiotardivo.github.io>
🌐 <https://www.linkedin.com/in/fabiotardivo>

FABIO TARDIVO

Curriculum Vitae

RESEARCH INTERESTS

Decision Diagrams · Constraint Programming · GPU Computing · Combinatorial Optimization · Neurosymbolic AI

EDUCATION

Ph.D. in Computer Science 2018 – 2025

Institution New Mexico State University, Las Cruces, New Mexico, USA
Advisor Dr. Enrico Pontelli
Thesis *Effective and Transparent Use of GPUs in Constraint Solving*

M.Sc. in Computer Science 2015 – 2017

Institution University of Udine, Udine, Italy
Advisor Dr. Agostino Dovier
Thesis *Redesign a Constraint Solver on GPU*
Grade 110/110 cum laude

Erasmus Exchange Program 2013

Institution Norwegian University of Science and Technology, Trondheim, Norway

B.Sc. in Computer Science 2008 – 2011

Institution University of Udine, Udine, Italy
Advisor Dr. Alberto Policriti
Co-advisor Dr. Cristian Del Fabbro
Thesis *Developing a distributed algorithm for the alignment of biological sequences*

WORK EXPERIENCE

Postdoctoral Researcher 2025 – 2026

Employer University of Connecticut, Storrs, Connecticut, USA
Supervisor Dr. Laurent Michel
Description Research on GPU-accelerated Decision Diagram-based optimization.
Description Mentoring Ph.D. students on adjacent research projects.

Research Assistant 2018 – 2025

Employer New Mexico State University, Las Cruces, New Mexico, USA
Supervisor Dr. Enrico Pontelli
Description Research on GPU-accelerated constraint solving.
Description Investigated on-edge machine learning deployment on satellite hardware.

Web Developer 2020

Employer New Mexico State University, Las Cruces, New Mexico, USA
Description Developed the web interface for the university's COVID-19 contact tracing application.

Software & Firmware Developer 2014 – 2015

Employer Prosa S.r.l., Venice, Italy
Description Developed firmware and web interfaces for automotive and vending machine systems.

ACADEMIC CONTRIBUTIONS

Conference Presentations

CP – Intl. Conf. on Principles and Practice of Constraint Programming	2024
CPAIOR – Intl. Conf. on Integration of CP, AI, and Operations Research	2023
CILC – Italian Conf. of Computational Logic	2022
PADL – Intl. Symp. on Practical Aspects of Declarative Languages	2022

	PADL – Intl. Symp. on Practical Aspects of Declarative Languages	2021
	ICLP – Intl. Conf. on Logic Programming	2019
Teaching & Mentorship		
Role	Guest Lecture	2026
Institution	University of Connecticut, Storrs, Connecticut, USA	
Topic	GPU-accelerated Decision Diagram-based optimization	
Role	Thesis Co-advisor (with Dr. Antonella Meneghetti)	2025
Institution	University of Udine, Udine, Italy	
Thesis	<i>Renewable Energy Integration for Industry 5.0</i> (by Simone Bellan, M.Sc.)	
Role	Guest Lecture	2023
Institution	New Mexico State University, Las Cruces, New Mexico, USA	
Topic	GPU Programming	
Role	Substitute Teacher	2018
Institution	Raimondo D'Aronco State High School, Gemona del Friuli, Udine, Italy	
Topics	Electronics, Programming, Networking	
Peer Review		
Role	Sub-reviewer	
Journals	Intelligenza Artificiale (Italian Journal on Artificial Intelligence)	2026
	Constraints	2024
Conferences	CP – Intl. Conf. on Principles and Practice of Constraint Programming	2026
	CPAIOR – Intl. Conf. on Integration of CP, AI, and Operations Research	2026
	CPAIOR – Intl. Conf. on Integration of CP, AI, and Operations Research	2025
	AAMAS – Intl. Conf. on Autonomous Agents and Multiagent Systems	2023
	CILC – Italian Conf. of Computational Logic	2023
	ECAI – European Conf. on Artificial Intelligence	2020
	ICLP – Intl. Conf. on Logic Programming	2020
	CP – Intl. Conf. on Principles and Practice of Constraint Programming	2020
	LPNMR – Intl. Conf. on Logic Programming and Non-monotonic Reasoning	2019
	CP – Intl. Conf. on Principles and Practice of Constraint Programming	2019

PUBLICATIONS

- [1] F. Tardivo, L. Michel, and W.-J. van Hoeve, “GPU-Accelerated State Exploration for Anytime Exact Decision-Diagram Search,” in *Integration of Constraint Programming, Artificial Intelligence, and Operations Research - 23th International Conference, CPAIOR 2026, Rabat, Morocco, May 26 - May 29, 2026, Proceedings (To Appear)*, in Lecture Notes in Computer Science. Springer, 2026.
- [2] E. Santi, A. Dovier, A. Formisano, and F. Tardivo, “GPU Accelerated Compact-Table Propagation,” *Theory Pract. Log. Program.*, vol. 25, no. 4, pp. 756–774, 2025, doi: 10.1017/S1471068425100197.
- [3] S. Travasci, F. Tardivo, and A. Formisano, “GPU-Accelerated Propagation for the Stable Marriage Constraint,” in *Proceedings of the 40th Italian Conference on Computational Logic, Alghero, Italy, June 25-27, 2025*, D. Guidotti, L. Pandolfo, and L. Pulina, Eds., in CEUR Workshop Proceedings. CEUR-WS.org, 2025. [Online]. Available: <https://ceur-ws.org/Vol-4003/paper14.pdf>
- [4] E. Santi, F. Tardivo, A. Dovier, and A. Formisano, “GPU Accelerated Compact-Table Propagation,” *CoRR*, 2025, doi: 10.48550/ARXIV.2507.18413.
- [5] F. Tardivo, A. Dovier, A. Formisano, L. Michel, and E. Pontelli, “Constraint propagation on GPU: a case study for the cumulative constraint,” *Constraints An Int. J.*, vol. 29, no. 1–2, pp. 192–214, 2024, doi: 10.1007/S10601-024-09371-W.
- [6] F. Tardivo, L. Michel, and E. Pontelli, “CP for Bin Packing with Multi-Core and GPUs,” in *30th International Conference on Principles and Practice of Constraint Programming, CP 2024, Girona, Spain, September 2-6, 2024*, P. Shaw, Ed., in LIPIcs. Schloss Dagstuhl - Leibniz-Zentrum für Informatik, 2024, pp. 28:1–28:19. doi: 10.4230/LIPICS.CP.2024.28.
- [7] F. Tardivo, L. Michel, and E. Pontelli, “Constraint Propagation on GPU: A Case Study for the Bin Packing Constraint,” *CoRR*, 2024, doi: 10.48550/ARXIV.2402.14821.
- [8] F. Tardivo, A. Dovier, A. Formisano, L. Michel, and E. Pontelli, “Constraint propagation on GPU: A case study for the AllDifferent constraint,” *J. Log. Comput.*, vol. 33, no. 8, pp. 1734–1752, 2023, doi: 10.1093/LOGCOM/EXAD033.

- [9] F. Tardivo, A. Dovier, A. Formisano, L. Michel, and E. Pontelli, "Constraint Propagation on GPU: A Case Study for the Cumulative Constraint," in *Integration of Constraint Programming, Artificial Intelligence, and Operations Research - 20th International Conference, CPAIOR 2023, Nice, France, May 29 - June 1, 2023, Proceedings*, A. A. Ciré, Ed., in Lecture Notes in Computer Science. Springer, 2023, pp. 336–353. doi: 10.1007/978-3-031-33271-5_22.
- [10] F. Tardivo, A. Dovier, A. Formisano, L. Michel, and E. Pontelli, "Constraints propagation on GPU: A case study for AllDifferent," in *Proceedings of the 37th Italian Conference on Computational Logic, Bologna, Italy, June 29 - July 1, 2022*, R. Calegari, G. Ciatto, and A. Omicini, Eds., in CEUR Workshop Proceedings. CEUR-WS.org, 2022, pp. 61–74. [Online]. Available: https://ceur-ws.org/Vol-3204/paper_6.pdf
- [11] F. Tardivo and E. Pontelli, "Parallel Declarative Solutions of Sequencing Problems Using Multi-valued Decision Diagrams and GPUs," in *Practical Aspects of Declarative Languages - 24th International Symposium, PADL 2022, Philadelphia, PA, USA, January 17-18, 2022, Proceedings*, J. Cheney and S. Perri, Eds., in Lecture Notes in Computer Science. Springer, 2022, pp. 191–207. doi: 10.1007/978-3-030-94479-7_13.
- [12] F. Tardivo, L. Pham, T. C. Son, and E. Pontelli, "A Logic Programming Approach to Regression Based Repair of Incorrect Initial Belief States," in *Practical Aspects of Declarative Languages - 23rd International Symposium, PADL 2021, Copenhagen, Denmark, January 18-19, 2021, Proceedings*, J. F. Morales and D. A. Orchard, Eds., in Lecture Notes in Computer Science. Springer, 2021, pp. 73–89. doi: 10.1007/978-3-030-67438-0_5.
- [13] M. D. Bortoli, F. Igne, F. Tardivo, P. Totis, A. Dovier, and E. Pontelli, "Towards Distributed Computation of Answer Sets," in *Proceedings of the 34th Italian Conference on Computational Logic, Trieste, Italy, June 19-21, 2019*, A. Casagrande and E. G. Omodeo, Eds., in CEUR Workshop Proceedings. CEUR-WS.org, 2019, pp. 316–326. [Online]. Available: <https://ceur-ws.org/Vol-2396/paper36.pdf>
- [14] F. Tardivo, "Experimenting with Constraint Programming on GPU," in *Proceedings 35th International Conference on Logic Programming (Technical Communications), ICLP 2019 Technical Communications, Las Cruces, NM, USA, September 20-25, 2019*, B. Bogaerts, E. Erdem, P. Fodor, A. Formisano, G. Ianni, D. Inclezan, G. Vidal, A. Villanueva, M. D. Vos, and F. Yang, Eds., in EPTCS. 2019, pp. 427–432. doi: 10.4204/EPTCS.306.58.
- [15] C. D. Fabbro, F. Tardivo, and A. Policriti, "A Parallel Algorithm for the Best k-Mismatches Alignment Problem," in *22nd Euromicro International Conference on Parallel, Distributed, and Network-Based Processing, PDP 2014, Torino, Italy, February 12-14, 2014*, IEEE Computer Society, 2014, pp. 586–589. doi: 10.1109/PDP.2014.51.